T.I.P. TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE027	**SECTION:**	CPE41S2
DATE PERFORMED:	24 July 2026	**DATE SUBMITTED:**	24 July 2026
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ACTIVITY NO. 3 Statistical Analysis of Signals

1. PROCEDURE OUTPUT

GDrive for csv files: (https://drive.google.com/drive/folders/1rBRuSPATvIe8h6--bhXZvys1en3AEE_G)
Github: https://github.com/leon-pscl/CPE027-pscl/blob/main/HOA3/Pascual_HOA3_CPE027_CPE41S2.ipynb
A sensor managed to sample data in 0 to 195 steps. You will be given access to a repository of each sample in the form of CSV files. Use only csv files coming from machine B1. There is a total of 196 sample files, each of those files contain 41,666 data points. Disregard the file labeled: Base Noise.

1. Create a program that can calculate the mean, standard deviation, and standard error of voltage and resistance for **each** STEP, and consolidate it in a separate (new) csv file.
   - *First, the data undergoes preprocessing to be able to use only the files required. This involved creating a list that gets only the CSV files related to Machine B1, excluding the BaseNoise files as well. In total, this would get 196 CSV files*
   - *Since each file represents one step, make a loop to read each csv file and compute for the mean, standard deviation, and standard error of voltage and resistance for each. This loop does it by first extracting the voltage and resistance columns from the dataframe Then, the variables needed for computation are extracted:*
     - *sample size (n)*
     - *delta degrees of freedom (since we're using sample standard deviation, ddof = 1)*
     - *mean*
     - *standard deviation*
     - *standard error (standard deviation divided by square root of the sample size)*

   The loop then stores the values computed, contained in a dataframe, then saved as a CSV file

```
Processed 196 steps. Saved to step_statistics.csv
Out[19]: (196, 7)

In [22]: stats_df.head(10)

Out[22]:
```

	Step	Voltage_Mean	Voltage_Std	Voltage_StdErr	Resistance_Mean	Resistance_Std	Resistance_StdErr
0	0	-0.000211	0.064777	0.000317	0.0	0.0	0.0
1	1	-0.000698	0.063847	0.000313	0.0	0.0	0.0
2	2	-0.000251	0.064671	0.000317	0.0	0.0	0.0
3	3	-0.000215	0.064206	0.000315	0.0	0.0	0.0
4	4	-0.000307	0.063391	0.000311	0.0	0.0	0.0
5	5	-0.000262	0.064165	0.000314	0.0	0.0	0.0
6	6	-0.000253	0.063858	0.000313	0.0	0.0	0.0
7	7	-0.000243	0.064385	0.000315	0.0	0.0	0.0
8	8	-0.000271	0.063752	0.000312	0.0	0.0	0.0
9	9	-0.000185	0.064934	0.000318	0.0	0.0	0.0

For additional information, the rows with the highest values in terms of Voltage_Mean, Voltage_Std, and Voltage_StdErr were extracted

```
Top 5 Highest Voltage Mean:
```

	Step	Voltage_Mean	Voltage_Std	Voltage_StdErr	Resistance_Mean	Resistance_Std	Resistance_StdErr
127	127	-0.000061	0.150408	0.000737	0.0	0.0	0.0
93	93	-0.000085	0.066025	0.000323	0.0	0.0	0.0
113	113	-0.000096	0.279396	0.001369	0.0	0.0	0.0
128	128	-0.000096	0.152272	0.000746	0.0	0.0	0.0
64	64	-0.000097	0.064632	0.000317	0.0	0.0	0.0

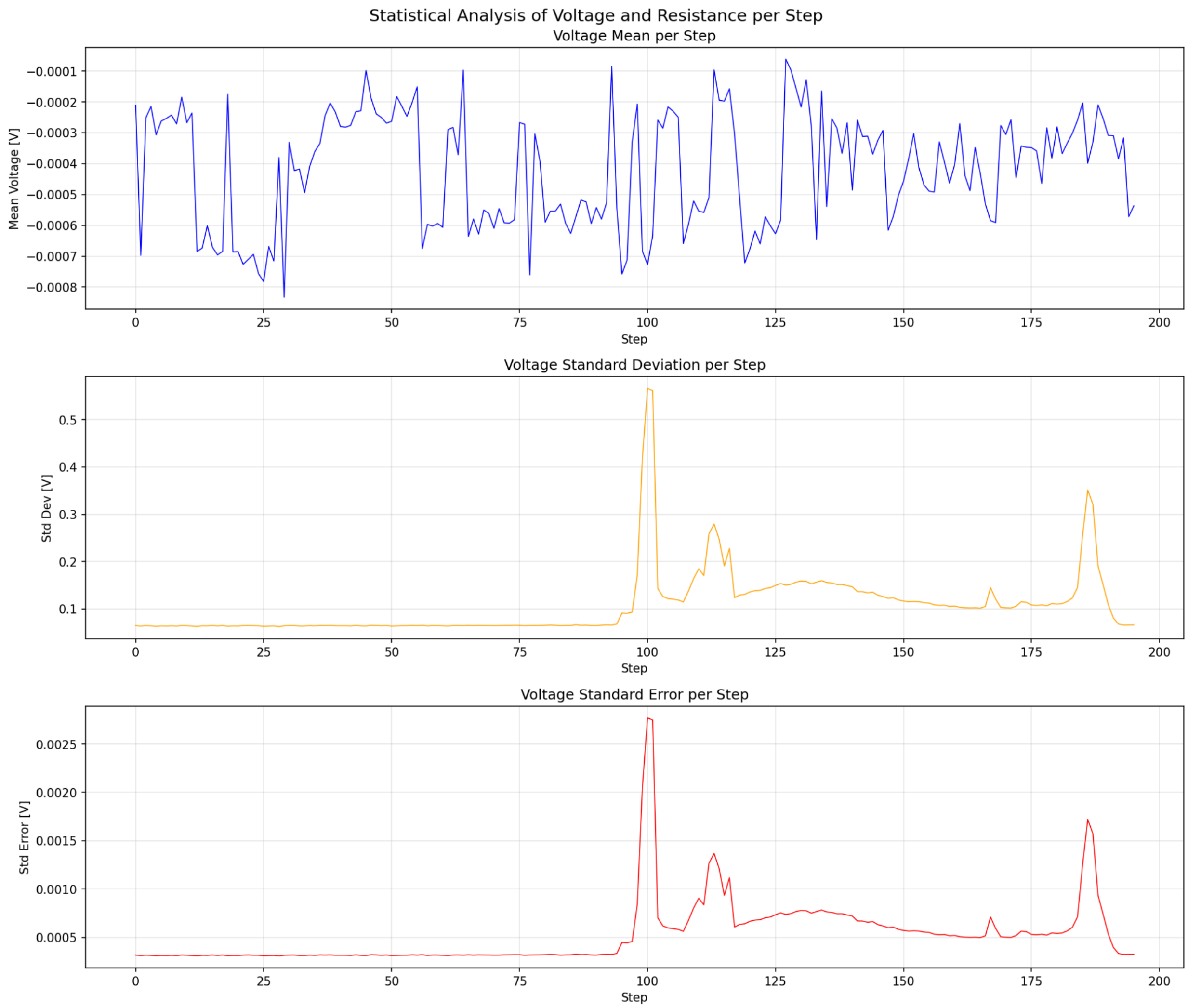
```
Top 5 Highest Voltage Standard Deviation:
```

	Step	Voltage_Mean	Voltage_Std	Voltage_StdErr	Resistance_Mean	Resistance_Std	Resistance_StdErr
100	100	-0.000727	0.565613	0.002771	0.0	0.0	0.0
101	101	-0.000634	0.561097	0.002749	0.0	0.0	0.0
99	99	-0.000684	0.419074	0.002053	0.0	0.0	0.0
186	186	-0.000399	0.351269	0.001721	0.0	0.0	0.0
187	187	-0.000330	0.321054	0.001573	0.0	0.0	0.0

```
Top 5 Highest Voltage Standard Error:
```

	Step	Voltage_Mean	Voltage_Std	Voltage_StdErr	Resistance_Mean	Resistance_Std	Resistance_StdErr
100	100	-0.000727	0.565613	0.002771	0.0	0.0	0.0
101	101	-0.000634	0.561097	0.002749	0.0	0.0	0.0
99	99	-0.000684	0.419074	0.002053	0.0	0.0	0.0
186	186	-0.000399	0.351269	0.001721	0.0	0.0	0.0
187	187	-0.000330	0.321054	0.001573	0.0	0.0	0.0

2. Then plot the resulted new csv file that shows the mean, standard deviation, and standard error of voltage and resistance for **each** STEP



It can be observed that the mean voltage across each step fluctuates quite randomly, with the highest mean voltage being recorded at step 127.

Interestingly, the standard deviation and standard error of Machine B1 had identical behavior. The values are quite stationary up until around the step 95 mark, signifying that there may have been factors that introduced changes to the system starting from that period.

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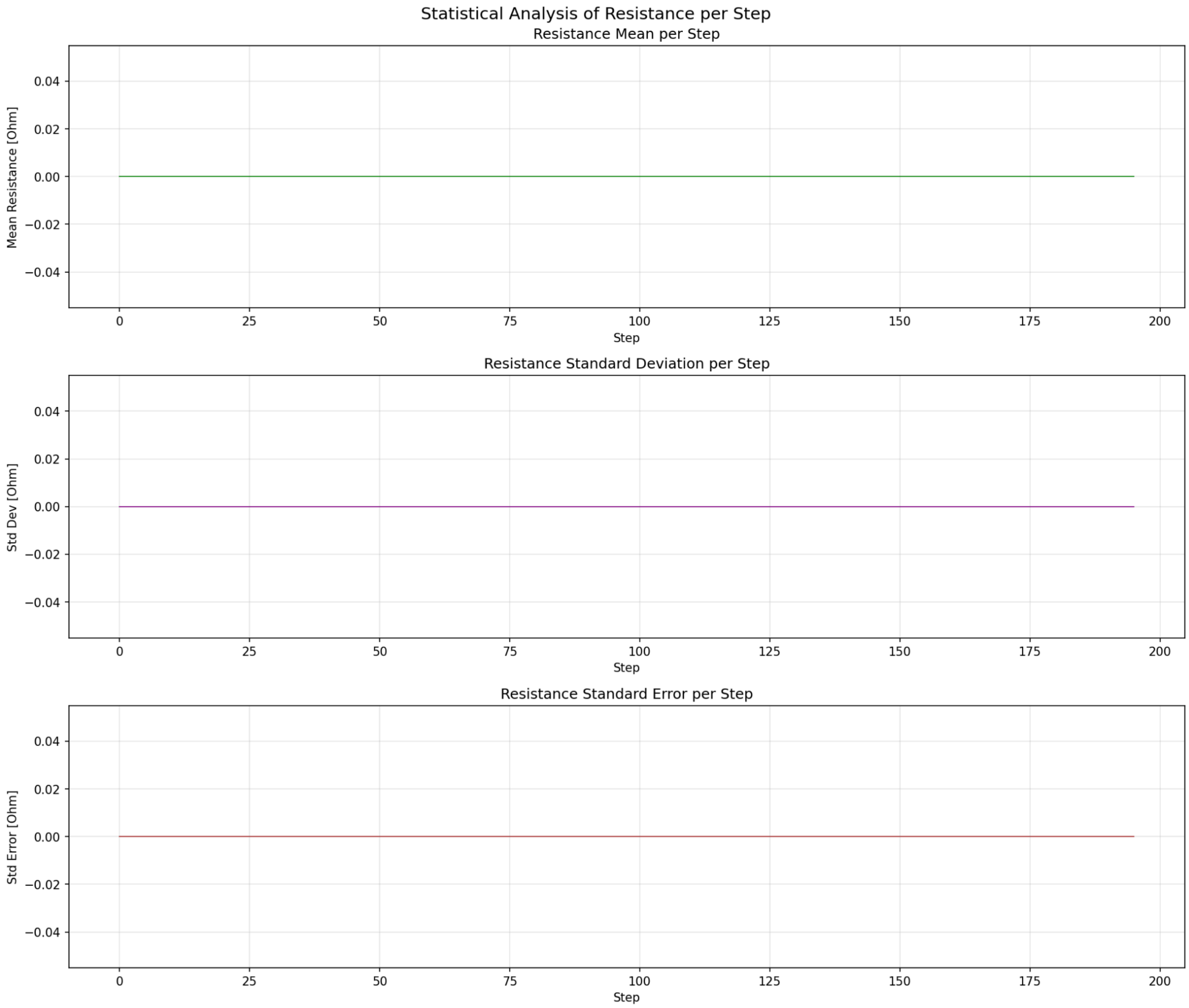
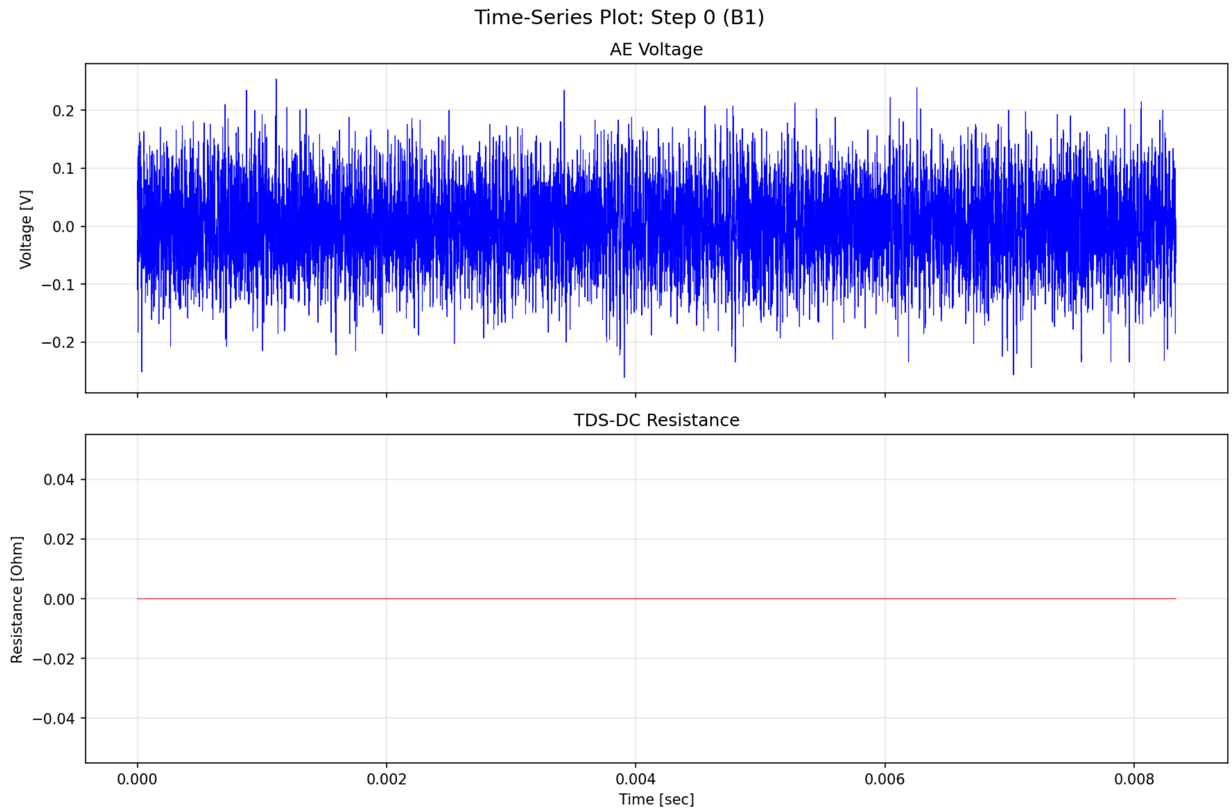
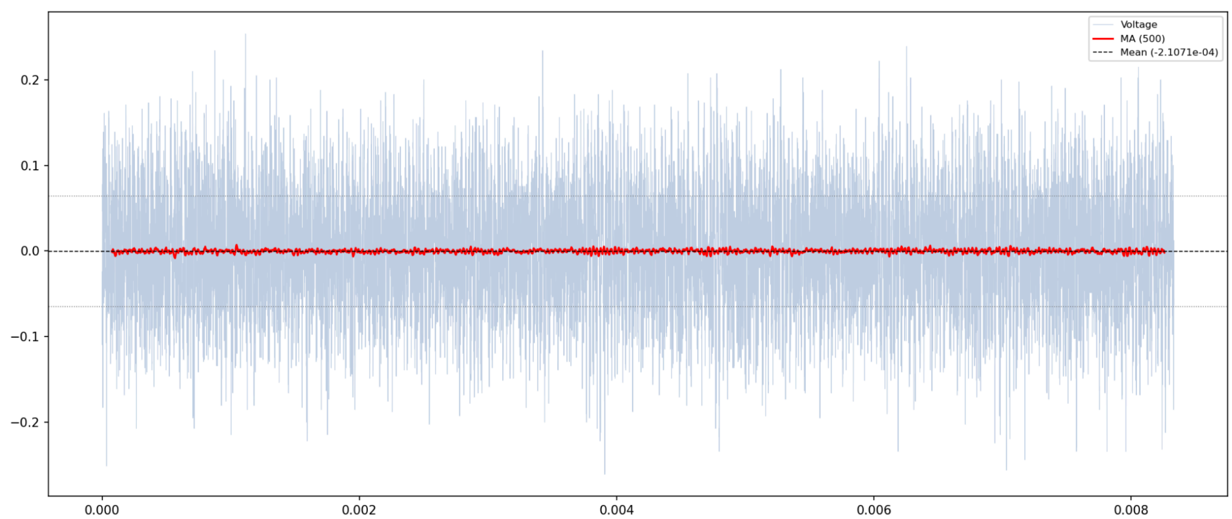


The recorded resistance across all steps for the machine is 0 all throughout. Nothing significant can be inferred here.

3. Create a program that is able to plot the time-series plots of
G1_Hd1_FTDS_SP1_050418_162003_1_0_0_B1.csv

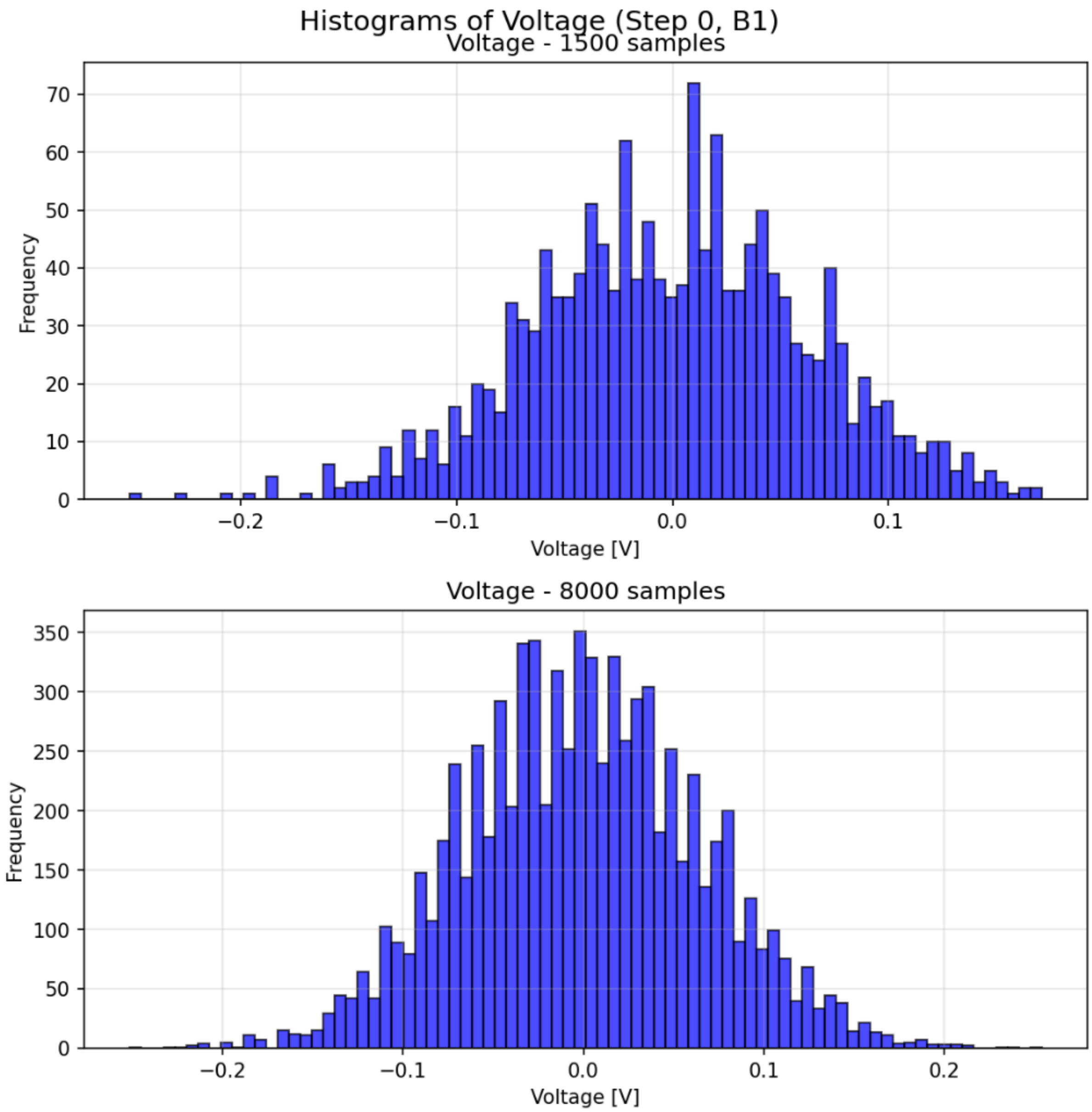


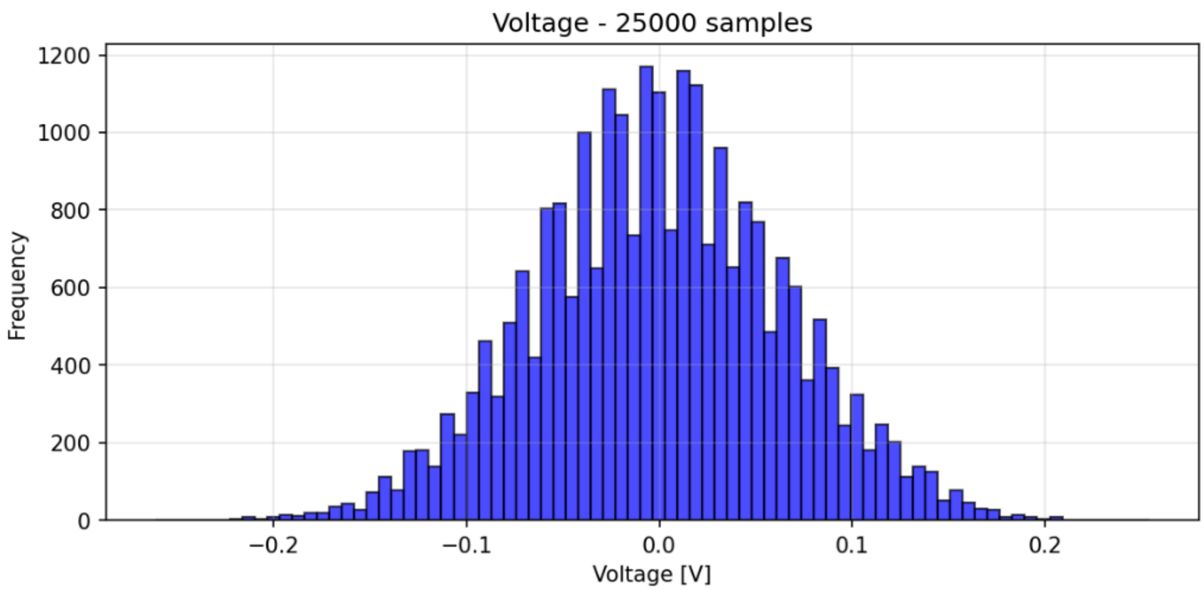
The current visualizations show that the recorded voltages within the initial step are quite noisy and highly varied. We can use a moving average to look for trends amidst the highly volatile data.

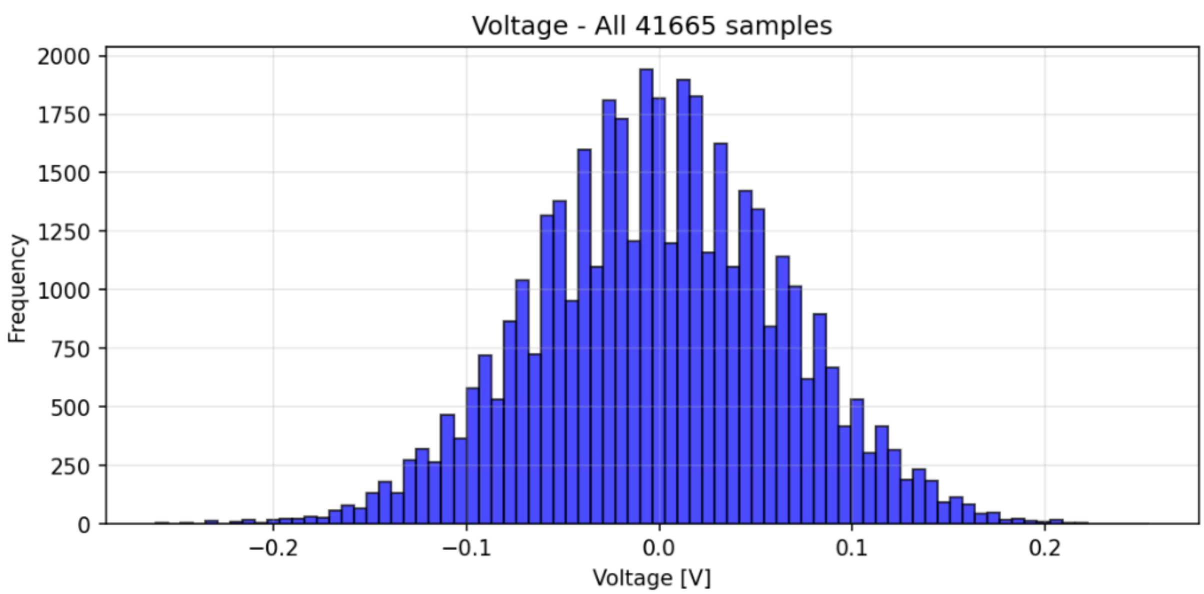


In getting the moving average per 750 samples (window = 750), it can be observed that the signals are quite stationary, and there are no notable anomalies within the machine's records. It suggests that the machine runs as usual, despite the volatility of the data.

3. Also, plot the histogram of voltage and resistance of **G1_Hd1_FTDS_SP1_050418_162003_1_0_0_B1.csv** for its first 1500 samples. Then try 8000 samples, then 25000 samples, and lastly, all samples.







From the histograms, it can be observed that the recorded samples are quite consistent with a normal distribution curve. The difference is that in the first 1500 samples, the distribution is left-skewed, suggesting that the samples are skewed towards the slightly higher voltages. However, as more samples get included in the histograms, the skewness slowly disappears.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

In summary, it can be concluded that the recorded voltage samples from Machine B1 show no major anomalies, as shown by the moving average. However, it should be noted that a sudden spike in standard deviation and standard error around step 95 onwards show that there may have been variables that must have caused it, and should be a case for inquiry. In addition to that, the histograms show skewness towards the higher voltages in the first steps and initial samples. This skewness disappears as more samples get taken into consideration.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Github Copilot, Opencode, Claude**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code Debugging, Code assistance, Code explanation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was used to help understand how the tasks needed to be done can be translated into readable, functional code. The methods in data preprocessing and data interpretation were done with my own effort and analysis.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.